The asymptotic Genius Conjecture

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Abstract

For every fixed index i, we express $F_i - u_i$ as a finite sum of monomials with rational coefficients in p, each of order $O(p^{-1})$. This proves Federbush's second Genius Conjecture. We also determine the first correction explicitly.

Task 1. Full solution

Statement and coefficient formula

Use the setting of Sections 1–2 of [1]:

$$U(x) = \sum_{k=1}^{p} u_k x^k, \quad u_1 = 1, \qquad D(x) = \sum_{k=1}^{p} d_k x^k, \qquad A(x) = e^{D(x)} = \sum_{k\geq 0} a_k x^k, \quad a_p = 0.$$

The functions $F_1 = 1, F_2, \ldots, F_p$ are uniquely specified by

$$[x^p]A(x)e^{yU(x)} = [t^p]\exp\left(y\sum_{k=1}^{p} F_k t^k\right). \tag{1}$$

Conjecture 2 asks for a finite monomial expansion of $F_i - u_i$ whose coefficients are rational functions of p tending to zero. As usual for these indexed rational functions, the limit keeps i fixed and takes integers $p \geq i$ to infinity.

Theorem 1. *For every fixed $i \geq 2$ and all integers $p \geq i$,*

$$F_i - u_i = \sum_{k=1}^{i-1}\sum_{m=1}^{i-k}\sum_{\substack{\ell_1+\cdots+\ell_m=i-k\\ \ell_\nu\geq 1}} \frac{k\prod_{j=1}^{m-1}(i-jp)}{m!\prod_{\nu=1}^{m}(p-\ell_\nu)} u_k \prod_{\nu=1}^{m} a_{\ell_\nu}. \tag{2}$$

Every rational coefficient in this expression is $O(p^{-1})$. After expanding the a_ℓ into monomials in the d_k, this proves Conjecture 2. More precisely, coefficientwise in the variables,

$$F_i = u_i + \frac{1}{p}\sum_{k=1}^{i-1} k u_k d_{i-k} + O(p^{-2}). \tag{3}$$

A formal coordinate formula

For a self-contained proof, we recall the coordinate construction used for the companion linearity result [2]. Put

$$W_p(x) = \sum_{k=0}^{p-1} \frac{p}{p-k} a_k x^k, \qquad \phi(x) = x W_p(x)^{-1/p}, \qquad \psi = \phi^{-1}.$$

Fractional powers have constant term 1, and $\phi = x + O(x^2)$ has a unique formal compositional inverse. Differentiation gives

$$\frac{\phi'}{\phi^{p+1}} = x^{-p-1}\left(W_p - \frac{x}{p} W_p'\right) = x^{-p-1} \sum_{k=0}^{p-1} a_k x^k.$$

Since $a_p = 0$, formal residue substitution yields, for every power series H,

$$[x^p] A(x) H(x) = \operatorname{Res}_x \frac{H(x)\phi'(x)}{\phi(x)^{p+1}} = [t^p] H(\psi(t)).$$

Apply this to $H = e^{yU}$. It follows that $F_i = [t^i] U(\psi(t))$. Uniqueness can also be seen directly: the coefficient of y^{p-i+1} in (1) determines F_i with multiplier $1/(p-i)!$ once the lower F_k have been determined. Formal integration by parts now gives

$$F_i = \operatorname{Res}_x \frac{U(x)\phi'(x)}{\phi(x)^{i+1}} = \frac{1}{i}[x^{i-1}] U'(x) W_p(x)^{i/p} = \sum_{k=1}^{i} \frac{k}{i} u_k [x^{i-k}] W_p(x)^{i/p}. \tag{4}$$

These are standard formal residue and Lagrange–Bürmann operations; no analytic limit or convergence assumption is involved.

Finite rational coefficients and their limits

In (4), the term $k = i$ is exactly u_i. For $k < i$, expand the binomial series

$$W_p^{i/p} = \sum_{m \geq 0} \binom{i/p}{m} \left(\sum_{\ell=1}^{p-1} \frac{p}{p-\ell} a_\ell x^\ell\right)^m.$$

Only $1 \leq m \leq i-k$ and positive compositions $\ell_1 + \cdots + \ell_m = i-k$ contribute. Their scalar coefficient is

$$\frac{k}{i}\binom{i/p}{m} \prod_{\nu=1}^{m} \frac{p}{p-\ell_\nu} = \frac{k \prod_{j=1}^{m-1}(i-jp)}{m! \prod_{\nu=1}^{m}(p-\ell_\nu)},$$

proving (2). Its numerator has degree at most $m-1$ in p and its denominator has degree m. Also $\ell_\nu \leq i-1 < p$, so no denominator vanishes in the allowed range. Thus every displayed coefficient is a rational function that is $O(p^{-1})$.

For each $\ell \leq i-1$, the coefficient a_ℓ is the finite polynomial

$$a_\ell = \sum_{\substack{b_1,\ldots,b_\ell \geq 0 \\ b_1 + 2b_2 + \cdots + \ell b_\ell = \ell}} \prod_{s=1}^{\ell} \frac{d_s^{b_s}}{b_s!}. \tag{5}$$

Substituting these polynomials into (2) gives a finite list of monomials $u_k \prod d_s^{b_s}$. The list depends on i, but not on $p \geq i$. Each coefficient is a rational constant times one of the rational functions above, or a finite sum of them, so each tends to zero. This is precisely the asserted monomial decomposition. Notice that d_p does not occur: the constraint $a_p = 0$ therefore imposes no extra restriction on the lower variables used in this formula.

The first correction

For a fixed composition in (2), its scalar coefficient has expansion

$$\frac{k \prod_{j=1}^{m-1}(i - jp)}{m! \prod_{\nu=1}^{m}(p - \ell_\nu)} = \frac{(-1)^{m-1}k}{mp} + O(p^{-2}).$$

For $s = i - k \geq 1$, the finite sum of the leading terms in the a_ℓ is

$$\sum_{m=1}^{s} \frac{(-1)^{m-1}}{m} \sum_{\substack{\ell_1 + \cdots + \ell_m = s \\ \ell_\nu \geq 1}} \prod_{\nu=1}^{m} a_{\ell_\nu} = [x^s] \log A(x) = d_s.$$

This proves (3) and completes the theorem. □

Verification and scope

Four exact verification groups check the original identity, finite rational formula, formal expansion through second order, and explicit F_2, F_3 formulas. Their 255 cases use exact fractions and supplement the proof for arbitrary fixed i.

All six pages of the primary were read. Conjecture 2 is distinct from the linearity statement in Conjecture 1, treated in [2]. The setting is Federbush's; the Bell formulation is credited there to David Williams. Bounded checks found no journal final or prior external full target claim as of September 18, 2026. Review is by the authoring agent only, without independent peer review or proof-assistant certification. The limit is coefficientwise at fixed i.

Tasks 2–5. Proposed follow-up directions

Task 2. Derive a general closed recurrence for all coefficients of the fixed-index expansion of F_i in inverse powers of p.

Task 3. Establish useful estimates when i also grows with p, including the regime $i/p \to \alpha \in (0, 1)$.

Task 4. Classify cancellations among the rational coefficients in (2), and determine minimal common denominators.

Task 5. Apply the exact transformation to the auxiliary coefficient corrections in the graph-positivity setting, with explicit dependence on the degree and perturbation parameters.

References

[1] P. Federbush, *The Genius Conjectures (via Bell Polynomials)*, arXiv:2002.03814v1, 2020. Conjecture 2, page 2.

[2] Principia Math In Progress, *Linearity in Federbush's Genius Conjecture*, MathDB 345451 solution package, September 18, 2026. The coordinate argument is reproduced above.